👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 10 · STANDARD MPP.4

Math is Ingenuity

Lesson 10-4 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use ratios and models to explain how the Penny Farthing and gear-driven bicycles were ingenious solutions to a real problem.

Language Objective: I can explain a gear ratio using the words rotation, teeth, and 'for each ___, the wheel turns ___.'

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

ingenuity · **rotation** · **gear** · **ratio** · **equivalent ratios**

Level 3: try the blanks from memory first, then check the bank.

1. Coming up with a different, sometimes unique solution is _____.

2. One full turn of a wheel, gear, or pedal is a _____.

3. A toothed wheel that passes motion through a chain is a _____.

4 A comparison of two quantities, like pedal turns to wheel turns, is a _____ .

5 Ratios that make the same comparison with different numbers are _____ _____ .

Write About the Math

How will your future vehicle be powered, and what math will be used in its design or use?
¿Con qué funcionará tu vehículo del futuro, y qué matemáticas se usarán en su diseño o su uso?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **ingenuity** — *ingenio*
- ☐ **rotation** — *rotación*
- ☐ **gear** — *engranaje*
- ☐ **ratio** — *razón*
- ☐ **equivalent ratios** — *razones equivalentes*

Model: *One pedal rotation makes one wheel rotation, so a bigger wheel's longer perimeter carries the bicycle farther with each pedal push.* — A strong answer connects the 1-to-1 ratio to the wheel's perimeter, explaining a bigger wheel travels farther per pedal push. A weak answer just says 'bigger is faster' without naming the ratio or perimeter.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Describe a future mode of individual transportation. A complete answer says how it is ___, names a ___ (or rate) that shapes it, and explains where that math appears in the vehicle's ___.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** ingenuity
2. **Notes 2:** rotation
3. **Notes 3:** gear
4. **Notes 4:** ratio
5. **Notes 5:** equivalent ratios
6. **Watch for this mistake:** *Thinking a rear gear with MORE teeth makes the wheel spin more — it is the opposite: the 32-tooth gear gives only 1.5 rotations per pedal rotation, while the 12-tooth gear gives 4.*

7. **Try It 1:**
8. **Try It 2:**